

TalentForge AI — Day-by-Day Execution Plan (AI-Powered Student Journey)

Team: Rohan (Backend · Sandbox · AI Services · Infra) · Karthikeyan (Frontend · Tools · AI UX · Deploy) | **Start:** Mon Jul 13, 2026 → **Launch:** Fri Aug 14, 2026

Workflow coverage (14 journey steps): Register **S1** · Profile **S2** · Domain **S3** · Baseline assessment + re-assess loop **S4** · AI learning path **S5** · Training modules **S6** · Challenges & work **S7-S8** · Recognition **S9** · Mentor feedback **S10-S11** · Talent profile **S12** · Recruiter visibility & shortlist **S13-S14** . **AI stack:** LLM API via backend proxy (provider-configurable) + pgvector embeddings in Postgres (no separate Pinecone). **Deferred post-launch:** XGBoost ranking, adaptive learning loop, mock video interviews (Video SDK), payment gateway, ECE domain.

■ Header ■ Day cell ■ Joint session **P0** critical path **AI** LLM / embedding feature **STRETCH** skippable **EXIT** weekly gate

WEEK 1 — Foundation, Onboarding & Domain Selection (Jul 13–17)

Day	Rohan — Backend / AI Services / Infra	Karthikeyan — Frontend / AI UX	Done When
Day 1 Mon Jul 13	- Monorepo: <code>npm init -w backend -w frontend -w worker</code>; Express + TS scaffold <code>docker-compose.yml</code>: postgres:16 with <code>pgvector/pgvector:pg16</code> image, redis:7, minio + healthchecks <code>schema.prisma</code>: User, StudentProfile (education, skills, certs, github, resumeS3Key), Domain, Problem, Submission, Assessment, LearningPath, Badge, Review, Shortlist - Migration incl. <code>CREATE EXTENSION vector</code>	- Vite + React 18 + TS scaffold; tailwind, TanStack Query, router, socket.io-client <code>AppShell.tsx</code> + design tokens - Register/Login UI S1 (react-hook-form + zod)	- Compose boots PG (with vector ext) + Redis + MinIO healthy <code>psql \dx</code> shows <code>vector</code> - Shell + auth pages render
Day 2 Tue Jul 14	<code>routes/auth.ts</code>: register (bcrypt), login → JWT 15m + refresh in Redis <code>SET refresh:{id} EX 604800</code> - Google OAuth (passport-google-oauth20); <code>GET /auth/me</code> - zod validation middleware on all bodies	- Axios instance + JWT interceptor + refresh-on-401 - MSW mocks for all planned endpoints - Auth flow wired; <code>RequireAuth.tsx</code> guard	- Register → login → <code>/auth/me</code> E2E S1 - OAuth round-trip OK; UI runs on mocks offline

Day	Rohan — Backend / AI Services / Infra	Karthikeyan — Frontend / AI UX	Done When
Day 3 Wed Jul 15	• Joint 2h: freeze <code>openapi.yaml</code> (~32 endpoints) + socket events <code>grading:*</code>, <code>copilot:*</code> • Profile API <code>S2</code> : CRUD education/skills/certs/links; resume upload → presigned S3 PUT • AI <code>POST /profile/parse-resume</code> : LLM proxy extracts skills/education JSON from resume text (pdf-parse) → prefill profile	• Joint 2h: contract freeze — regen MSW from spec • Profile wizard <code>S2</code> : 3-step form (basics → education/skills → GitHub/links), resume dropzone • “Parsed from resume — review & confirm” diff-style prefill UI	• Spec committed, both build against it • Upload resume → parsed skills prefill form → save profile
Day 4 Thu Jul 16	• Domain seed <code>S3</code> : AI/ML, Web Dev, Data Science, Cyber Security, Cloud — problems tagged <code>domainId</code> • <code>POST /profile/domain</code> {domainId, careerGoals} • Seed 3 problems/domain-agnostic (Two Sum, LRU Cache, Rate Limiter) with hidden test cases • BullMQ <code>grading</code> queue + worker skeleton emitting socket events	• Domain picker <code>S3</code> : card grid with icons, career-goals textarea • Monaco integration: <code>@monaco-editor/react</code>, language selector (py/js/java), boilerplate per problem, <code>Ctrl+Enter</code> • Problem board + detail (markdown statement, resizable split)	• Domain saved to profile; board filters by domain • Submit → dummy <code>grading:complete</code> in browser
Day 5 Fri Jul 17	• GitHub Actions CI (lint, typecheck, prisma validate, build) • Provision EC2 t3.medium; <code>docker-compose.prod.yml</code>; Caddy auto-HTTPS <code>staging.talentforge.in</code> • Secrets: <code>.env</code> on server only; LLM API key env var	• Wire full onboarding (register → profile → domain) on staging • Toasts, error boundary, 404, loading skeletons • CI deploys frontend to staging	• EXIT <code>S1</code>→<code>S3</code> onboarding + editor live on staging; CI green

WEEK 2 — Sandbox Grading & Baseline Assessment (Jul 20–24)

CRITICAL PATH

Day	Rohan — Backend / AI Services / Infra	Karthikeyan — Frontend / AI UX	Done When
Day 6 Mon Jul 20	• P0 Runner images: python:3.11-slim+pylint, node:18-alpine+eslint, temurin:17+checkstyle; non-root <code>runner</code> user • dockerode isolation: <code>NetworkMode: 'none'</code>, <code>Memory: 256MB</code>, <code>ReadOnlyRootfs</code>, tmpfs 50m, <code>PidsLimit: 64</code>, 30s kill timer	• Submit UX: spinner → live chips (Queued → Running → Graded) via <code>useGradingSocket(id)</code> • <code>ResultsPanel1.tsx</code> collapsible shell	• 3 languages execute; <code>curl</code> in container fails; infinite loop killed at 30s

Day	Rohan — Backend / AI Services / Infra	Karthikeyan — Frontend / AI UX	Done When
Day 7 Tue Jul 21	<code>grader/correctness.ts</code>: stdin per case, diff stdout, weighted pass % → 0–100; persist <code>scores.cases[]</code> - Java <code>javac</code> compile step → <code>COMPILE_ERROR</code> path	- Per-case table (pass/fail, expected-vs-actual diff, time, memory) - Animated score ring; <code>COMPILE_ERROR</code> / <code>TIMEOUT</code> / <code>OOM</code> states	Correct solution = 100 in all langs; failing cases show diffs
Day 8 Wed Jul 22	<code>grader/complexity.ts</code>: $n/2n/4n$ scaling fit → O-class vs <code>expectedComplexity</code> <code>grader/style.ts</code>: linters in-container → 0–100; composite <code>0.6/0.3/0.1</code>	- Score bars + composite AI Coaching feedback: <code>POST /feedback/format</code> LLM proxy → 3 actionable bullets, typing effect, flag <code>VITE_FF_LLM_FEEDBACK</code>	$O(n^2)$ scores below $O(n)$; AI bullets <3s or flag off cleanly
Day 9 Thu Jul 23	- Precheck blocklist (subprocess / <code>child_process</code> / <code>Runtime.exec...</code>) → <code>BLOCKED</code> pre-run; queue never stalls; retry infra ×2 only - Baseline assessment API S4: 40-item bank (aptitude, technical MCQ, logic, communication Likert); <code>POST /assessment/start submit</code>; score % + section breakdown; pass ≥70 else <code>FAILED</code> + <code>reassessAvailableAt</code> (48h) — the re-assess learning loop	- Assessment UI S4: timed 20-min flow, progress bar, <code>sessionStorage</code> autosave - Result screen: <code>PASSED</code> (green, % ring) / <code>FAILED</code> (re-assess countdown + “what to review” list) - Submission history page + resubmit cooldown chip	- Malicious code blocked pre-execution - Score 78% → <code>PASSED</code> unlocks challenges; <70% → re-assess loop visible
Day 10 Fri Jul 24	Joint E2E: 10 canned solutions × 3 langs; fix all - artillery: 20 concurrent submissions p95 <5s; Sentry node/worker	Joint E2E browser side; fix top-10 papercuts - Sentry React + replay; socket reconnect + 5s poll fallback	EXIT S4 gate + grade <5s p95 on staging; 20 concurrent, 0 failures

WEEK 3 — AI Layer: Learning Path, Copilot, Talent Profile, Mentor Review (Jul 27–31)

Day	Rohan — Backend / AI Services / Infra	Karthikeyan — Frontend / AI UX	Done When
Day 11 Mon Jul 27	- Badge service S9: score ≥75 → Badge (uuid verifyId, <code>AI_VERIFIED</code>); public <code>GET /verify/:id</code> - Certificate PDF (pdf-lib + QR) → S3; leaderboard query (XP = sum of badge scores)	- Badge celebration (confetti), gallery, public verify page + OG tags S9 - Leaderboard page (rank, badges, XP) — anonymized toggle	Badge auto-award works; verify URL public; certificate QR resolves; leaderboard ranks

Day	Rohan — Backend / AI Services / Infra	Karthikeyan — Frontend / AI UX	Done When
Day 12 Tue Jul 28	AI Learning path S5 : <code>POST /learning-path/generate</code> — LLM prompt with profile + domain + assessment sections → strict JSON (milestones[], weeklyGoals[], recommendedProblemSlugs[]); zod-validate LLM output, retry ×1 on invalid; persist; regenerate hook after each re-assessment	- Roadmap UI S5-S6 : vertical timeline (milestones → linked problem sets as “training modules”), checkbox progress, weekly goal cards “Regenerate path” with diff-style what-changed banner	- New user gets tailored roadmap referencing real seeded problems - Invalid LLM JSON never reaches DB (zod gate proven)
Day 13 Wed Jul 29	AI Copilot: <code>POST /copilot/chat</code> SSE streaming; context injection = profile + current page + last submission scores; modes mentor reviewer advisor via system prompt; Redis rate limit 30 msg/day/user; conversation table	Copilot drawer (global, all pages): streamed markdown, mode pill, suggested prompts per page (“Why did case 7 fail?” on results; “What should I learn next?” on roadmap)	Streaming answers grounded in the user’s own scores; rate limit returns friendly 429 at 31st msg
Day 14 Thu Jul 30	AI Talent profile S12 : skills-graph extraction — LLM maps profile + graded submissions → <code>SkillScore[]</code> rows; embedding job: profile text → vector via embeddings API → <code>profile_embedding vector(1536)</code> column + HNSW index - Nightly BullMQ repeatable job re-embeds changed profiles	- Talent profile page S12 : skills radar (Recharts), strengths/interests chips, achievements & scores timeline, GitHub link - Public/private toggle (<code>profilePublic</code>)	Profile shows extracted skills graph; <code>SELECT embedding <=> ...</code> returns sane neighbors
Day 15 Fri Jul 31	- Mentor review S10-S11 : <code>requireRole('REVIEWER')</code>; queue oldest-first; <code>POST /reviews/:id</code> {stars, comment, APPROVE REJECT} → EXPERT_VERIFIED or revoke + notify - Seed to 8 problems incl. flagship Load Balancer; nightly <code>pg_dump→S3</code>; prod env split	- Reviewer portal: queue, read-only Monaco, rating, approve/reject - Student notifications (bell) + mentor feedback shown on results S10 - LinkedIn badge share; landing page v1; onboarding tour	EXIT full journey S1→S12 on staging: onboard → assess → roadmap → solve → badge → mentor-verified → talent profile

WEEK 4 — Recruiter Matching, Hardening, Beta & Stretch (Aug 3–7)

Day	Rohan — Backend / AI Services / Infra	Karthikeyan — Frontend / AI UX	Done When
Day 16 Mon Aug 3	- Recruiter APIs S13–S14 : GET /candidates?minScore=&badge=&domain= (indexed); shortlist CRUD AI Smart matching: recruiter posts role text → embed → pgvector cosine ORDER BY embedding <=> \$1 LIMIT 20 → match % = normalized similarity × badge-score weight - Shortlist email alert to student (nodemailer / MSG91)	- Recruiter portal: register flow, Discover table (TanStack: match %, score sort, badge/domain filters) - Candidate drawer: skills radar, badges, best code sample, match explanation, Shortlist - Student: application status tracker (Shortlisted → Interview → Offer) S14	Role paste → ranked candidates with match %; shortlist notifies student; status visible both sides
Day 17 Tue Aug 4	P0 Beta launch — 15–25 students - Security: helmet, CORS allowlist, zod audit 100%, cookie flags, prompt-injection guard on copilot (system-prompt hardening + output length caps) - Live monitor Sentry + slow-query log; hotfixes	- Beta comms channel + in-app feedback widget → POST /feedback - Mobile pass (tables → cards; editor read-only banner on mobile) - Same-day UI hotfixes, GitHub beta issues	≥15 registered, ≥10 graded, ≥5 roadmaps generated; 0 open P1s
Day 18 Wed Aug 5	- k6 100 VUs (browse+submit+copilot mix); LLM cost guardrail: per-user daily token cap + cache identical roadmap prompts (Redis, 24h) - Compose restart:always, Redis AOF, disk alert; top-5 backend fixes	Joint 1h beta triage → ranked list; top-5 UX fixes - Lighthouse ≥90, LCP <1.5s (route code-split, font preload)	k6: error <1%, API p95 <200ms; LLM spend projection within budget; top-10 beta list closed
Day 19 Thu Aug 6	STRETCH Hardhat + OpenZeppelin TalentForgeBadge.sol ; deploy Polygon Amoy; POST /badges/:id/mint server-side signer → txHash If behind: SKIP — DB badges ship	STRETCH “Mint on-chain” button → PolygonScan link; hidden when VITE_FF_CHAIN=false - Recognition polish S9 : certificate download, rankings page final	Mint works on Amoy — or flag off with zero side effects
Day 20 Fri Aug 7	- RUNBOOK.md (+ LLM provider outage playbook: copilot/roadmap degrade gracefully, grading unaffected) - Backup restore drill; joint go/no-go	- Copy sweep, FAQ, 2-min demo video, recruiter one-pager PDF Joint go/no-go	EXIT restore drill OK; LLM-down mode verified; go/no-go = GO; 0 P1/P2

Day	Rohan — Backend / Infra	Karthikeyan — Frontend / Comms	Done When
Day 21 Mon Aug 10	CODE FREEZE bugfix-only - Prod regression: full S1→S14 journey ×3 languages; DNS + SSL on <code>app.talentforge.in</code>	Regression on Chrome/Firefox/Safari + 2 phones; OG previews validated	Journey checklist 100% green on production
Day 22 Tue Aug 11	SOFT LAUNCH ~50 invited students; watch Sentry, queue depth, LLM spend dashboard	Send invites; monitor first sessions (replay); fix onboarding friction same-day	≥30 signups, ≥15 graded, ≥20 roadmaps, 0 P1s
Day 23 Wed Aug 12	Fix soft-launch findings; UptimeRobot on <code>/health</code>, frontend, verify URL	Recruiter outreach: one-pager to 5–10 HR contacts; assist accounts; collect student quotes	≥3 recruiter accounts; ≥1 AI-matched shortlist made
Day 24 Thu Aug 13	Final backup + snapshot; alert thresholds (error rate, disk, queue, LLM spend); on-call plan	Launch assets final: founder post, college groups, Discord, demo video pinned	Posts scheduled; backups verified
Day 25 Fri Aug 14 	PUBLIC LAUNCH on-call: grading rate, uptime, queue, LLM latency/cost - Evening metrics snapshot for investor/customer deck	- Publish 10:00 IST; respond all day; hotfix UI - Evening “first verified talents” follow-up post	Targets: 100+ signups · 60+ assessments · 40+ roadmaps · 30+ submissions · 15+ badges · 3–5 recruiters · 1+ shortlist · 99%+ uptime · grading p95 <5s

Daily rhythm: 9:30 AM IST standup · PRs cross-reviewed <12h · Friday 4 PM demo + retro · staging auto-deploys on `main`, prod manual & tagged. **AI staging rule:** LLM features (parse, roadmap, copilot, embedding match) ship now; ML features needing training data (XGBoost ranking, adaptive learning) wait for post-launch pilot data. **Say no until Aug 15:** ECE domain, video interviews, payments, C++/Go/Rust.